

Ejercicio 1: $b = 5,5m$, $m = 1$, $n = 9,012$, $S_0 = 0,001$, $h_{ca} = 1,5m$

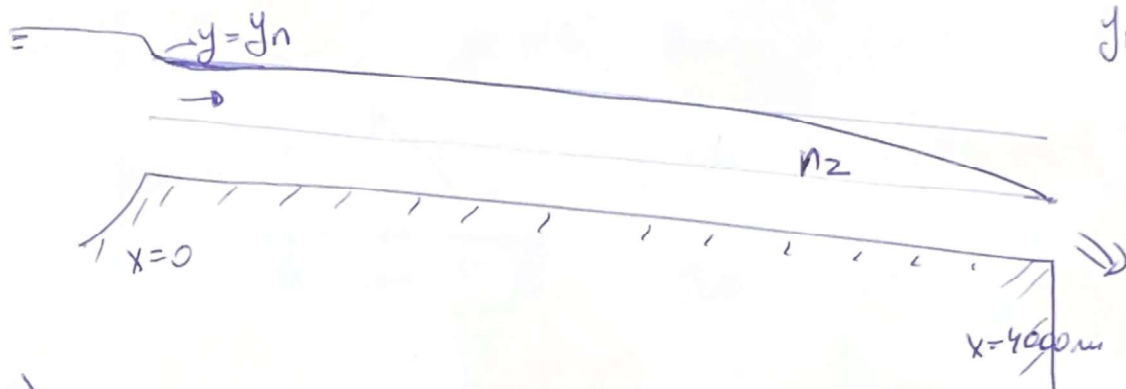
1) Supongo canal $n \rightarrow$ Filtro canal $\therefore \left. \begin{array}{l} E_1 = h_{ca} = 1,5m \\ y_1 = y_n \end{array} \right\}$

$\Rightarrow Q = 19,59 m^3/s$

$y_n = 1,19m$ $\rightarrow y_c = 1,02m \rightarrow y_n > y_c$
 vertico canal n

canal n caide libre $\rightarrow$ canal n_2 vertico

$y_{n2}(x=0) \approx y_n$



2) CONPUERTA en $x = 3200m$ (800m antes caide libre)

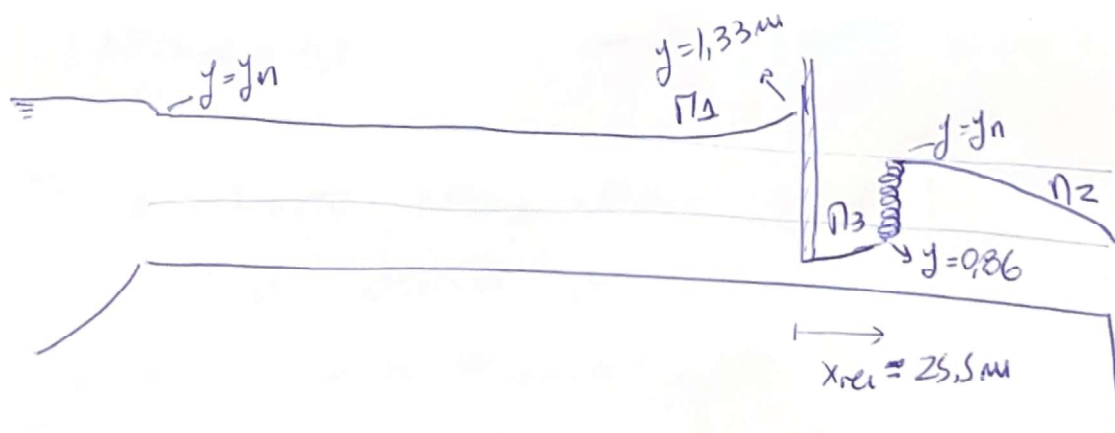
$a = 0,9m \rightarrow a < y_c$

$y_{n2}(x=3200) = 1,19m \rightarrow y_{n2}(\text{CONPUERTA})^* = 2,36m > a$

$\Rightarrow$ DESCARGA LIBRE

Punto n_3 n_2 ; equivarise carpente $y = a' = 1,33m > y_n$

calculo $y(x=0)$ DADO POR $n_2 \rightarrow y_{n2}(x=0) \approx y_n \rightarrow$ en caide n_1 $Q = 19,59 m^3/s$



3) con $a = 0,9m \rightarrow y_{n2}(\text{500m antes carpente}) = 1,2m$

Por lo que a $y_{n2}(\text{500m antes carpente}) = 1,8m$ (uso limite) $\rightarrow y_{en}$

carpente debe ser $2,27m$ (que es a') $\Rightarrow a = 0,547m$ $\sqrt{a = 0,55m}$

EJERCICIO 4:

$$1) H_m = H_B - H_A = H_{Ti} + \Delta H_i - (H_{Ts} - \Delta H_s)$$

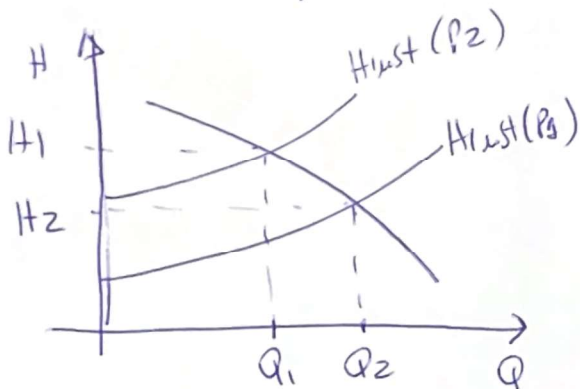
$$H_{Ti} = P_i / \gamma + z_i$$

$$H_{Ts} = z_s$$

$$\Delta H_i = \left(k_i + f_i \frac{L_i}{D_i} \right) \frac{Q^2}{2gA_i^2}$$

$$\Delta H_s = \left(k_s + f_s \frac{L_s}{D_s} \right) \frac{Q^2}{2gA_s^2}$$

Rango $P_i \rightarrow 1 \times 10^5 \text{ Pa}$ y $3 \times 10^5 \text{ Pa}$
(P_1) (P_2)



Rango Q (0,1023, 0,174) m^3/s

Rango H (40,10, 47,77) m

$$\text{Para } P_i = 1 \times 10^5 \text{ Pa} \rightarrow \begin{cases} Q = 0,174 \text{ m}^3/\text{s} \\ H = 40,10 \text{ m} \end{cases}$$

$$\text{Para } P_i = 3 \times 10^5 \text{ Pa} \rightarrow \begin{cases} Q = 0,1023 \text{ m}^3/\text{s} \\ H = 47,77 \text{ m} \end{cases}$$

$$2) Q = 0,11 \text{ m}^3/\text{s} \Rightarrow \text{en curva bombe} \rightarrow H_m \approx 47 \text{ m}$$

$$\Rightarrow f_i = f_s = 0,0153$$

$$\Rightarrow H_{Ti} = H_m - \Delta H_i + H_{Ts} - \Delta H_s = P_i / \gamma + z_i$$

$$\Rightarrow P_i = (47 + z_s - z_i - \Delta H_i - \Delta H_s) \cdot \gamma \approx \underline{\underline{2,02 \times 10^5 \text{ Pa}}}$$

$$NPSH_d = z_s - \frac{Q^2}{2gA_s^2} \left(f \frac{L_s}{D} + k_s \right) - z_B + 10,09 = 8,75 \text{ m}$$

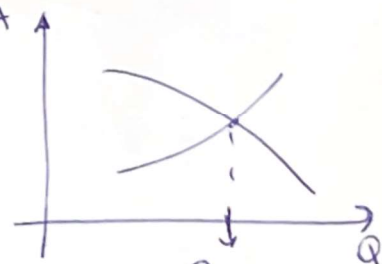
$$NPSH_{ref} \approx 9,24 \text{ m}$$

$\rightarrow$ NO CAVITA $NPSH_d > NPSH_{ref}$

$$3) \text{ CASO LIMITE } NPSH_d = NPSH_{ref}$$

$NPSH_d$ no depende de P_i

$$\rightarrow \text{Determino } Q_{\text{limite}} \Rightarrow Q_{\text{limite}} = 0,1198 \text{ m}^3/\text{s}$$



$$\rightarrow \text{Despego } P_i \text{ de la curva instalación sabiendo } H_{\text{en bombe}} \text{ y } f \text{ de la curva} \\ \text{en } (H = 46 \text{ m}, f = 0,0151) \Rightarrow \underline{\underline{P_{\text{limite}} = 2,54 \times 10^5 \text{ Pa}}}$$